

Vending Machine--- Communication Protocol

For Cash or Card Payment Integration

The communication method

UART serial communication, parameter configuration

Baud rate	9600bps
Data bit	8 bits
Stop bit	1 bit
Check digit	none
Flow control	none

VMC communication data frame format:

ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGTH	DATA	CHK
1 Byte	1 Byte	1 Byte	1 Byte	1 Byte	n Bytes	1 Byte

【Description】

- 1) vending machine controller
- 2) [Address] is fixed at 0xFF.
- 3) 【Frame number】 0x00.

[Data Header] When the Android APP is launched, it is 0x55, and when the vending machine device is launched, it is 0xAA;

- 4) 【Command】 See command details.
- 5) [Data length] 1 byte, representing the number of bytes occupied by the following data.
- 6) [Data] low position first, followed by DATA0, DATA1...DATAn;
- 7) [Verification] The verification mode is accumulation and verification: that is, the entire data frame is accumulated and all other bytes except [address] [frame number] [CHK], and the result of lower 8-bit data is taken;

Command details:

1. Android APP to get device ID

	ADDR	FRAME NUMBER	HEAD ER	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0x31	0x01	0xAD	CHK
VMC	0xFF	0x00	0xAA	0x31	0x0F	ID(15Bytes)	CHK

Note: VMC receives the command and responds

ID: 15 byte string data.

2. Control command

2-1 Delivery Command

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0x41	0x02	Cargo lane (1Byte) Delivery Quantity (1Byte)	CHK
VMC	0xFF	0x00	0xAA	0x41	0x02	Cargo lane (1Byte) delivery Quantity (1Byte)	CHK

【Note】 : VMC receives the instruction and responds, and returns to the original delivery instruction.

2-2 Remove fault command

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0xA2	0x01	0xFF	CHK
VMC	0xFF	0x00	0xAA	0xA2	0x01	0x01	CHK

【Note】 : VMC receives the command and responds, and VMC returns to the initial state.

2-3 Payment instructions

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
APP	0xFF	0x00	0x55	0x11	0x06	payment amount(4Bytes) payment method(1Byte) slot(1Byte)	CHK
VMC	0xFF	0x00	0xAA	0x11	0x01	0x01	CHK

【Note】 : VMC receives the instruction and responds, VMC waits for payment.

Payment amount: 4 bytes of data, value range 0-4294967295, amount = commodity price × 100

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value of 0 means cancel the payment.

value of 1 means offline coin waiting for payment.

value of 2 represents offline cashless waiting for payment.

value of 3 represents offline bill acceptor waiting to be paid.

slot: 1 byte of data, the value range is 1-255, representing the number of the slot waiting to be paid for delivery.

2-4 Coin change order

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
上位机	0xFF	0x00	0x55	0xB1	0x01	0xFF	CHK
VMC	0xFF	0x00	0xAA	0xB1	0x01	0x01	CHK

VMC receives the instruction and responds, and VMC returns coins.

2-5 Cashless payment cancellation transaction instruction (applicable to MDB non-cash payment peripherals)

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
App	0xFF	0x00	0x55	0xB2	0x01	0xFF	CHK
VMC	0xFF	0x00	0xAA	0xB2	0x01	0x01	CHK

【Note】 : VMC receives the instruction and responds, and VMC cancels the transaction.

2-6 Debit instruction (applicable to VMC payment panel)

CHK

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
上位机 App	0xFF	0x00	0x55	0xB3	0x06	金额(4Bytes)	CHK
VMC	0xFF	0x00	0xAA	0xB3	0x01	0x01	CHK

[Note]: VMC receives the command and responds, and VMC starts deduction:

Amount: 4 bytes of integer data, the value range is 0-4294967295, amount = deducted amount × 100.

2-7 Age recognition open command (applicable to peripherals with age recognition)

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
上位机	0xFF	0x00	0x55	0x12	0x01	年龄(1Byte)	CHK
VMC	0xFF	0x00	0xAA	0x12	0x01	0x01	CHK

【Note】 : VMC receives the instruction and responds, VMC starts age identification, and waits for the age verification of the certificate.

Age: 1 byte data, the value range is 1-99, representing the restricted age.

3. Status query

3-1 Query VMC status

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
App	0xFF	0x00	0x55	0xE1	0x02	slot(1Byte) Delivery quantity(1Byte)	CHK

VMC returns status

Delivery completed

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
VMC	0xFF	0x00	0xAA	0xE1	0x01	0x01	CHK

【Note】 : VMC receives the command and responds, and returns the delivery successfully.

Delivery failure

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
VMC	0xFF	0x00	0xAA	0xE1	0x01	error code(1Byte)	CHK

【Note】 : VMC receives the command and responds, and returns delively failure.

Fault Code: 1 byte of data.

0x02 : Freightway motor failure.

0x03: Device optical eye failure.

Payment completed

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
VMC	0xFF	0x00	0xAA	0xE1	0x04	金额(4Bytes) Amount (4Bytes)	CHK

[Note]: VMC receives the instruction and responds, and returns the payment amount:

4-byte integer data, value range 0-4294967295, amount = payment amount × 100.

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[Note]: When non-cash payment is used offline, the returned payment amount is equal to the product price × 100.

[Note]: When using coin/note payment offline, the returned payment amount is equal to the VMC balance.

3-2 Query VMC balance

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
App	0xFF	0x00	0x55	0xE1	0x04	0x00(4Bytes)	CHK
VMC	0xFF	0x00	0xAA	0xE1	0x04	Amount(4Bytes)	CHK

【Note】 : VMC receives the command and responds, and returns the balance.

1) Balance: 4 bytes of integer data, value range 0-4294967295, amount = VMC

3-3 Query the status of the coin change

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
App	0xFF	0x00	0x55	0x07	0x01	0x01	CHK
VMC	0xFF	0x00	0xAA	0x07	0x01	状态(1Byte)	CHK

【Note】 : VMC receives the command and responds, and returns to the status of the coin hopper.

Status 0: Represents that coins can be refunded.

1: Insufficient tokens, unable to refund.

3-4 Query age identification verification status (applicable to peripherals with age identification)

	ADDR	FRAME NUMBER	HEADER	CMD	DATA LENGHT	DATA	CHK
App	0xFF	0x00	0x55	0x06	0x01	0x01	CHK
VMC	0xFF	0x00	0xAA	0x06	0x01	状态(1Byte)	CHK

【Note】 : VMC receives the command and responds, and returns to the age verification status.

Status 0: means age verification failed.

1: means the age verification is successful.